

Midterm Exam

Time: 75 minutes**Total: 100 points****Name:** _____**Date:** _____

Instructions: This is a closed-book and closed-notes examination. No calculator is allowed. Answer all questions. For multiple-choice questions, select the **single best answer**. For short-answer questions, explain your reasoning clearly and concisely. Numerical calculations, where required, are deliberately simple and can be completed without a calculator.

Part I. Multiple Choice

40 points

Select the single best answer. Each question is worth 2 points.

1. What is the primary purpose of data preprocessing in machine learning?
 - A. To guarantee that a machine-learning model achieves high accuracy
 - B. To transform data into a clean and suitable form for analysis and modeling
 - C. To increase the number of observations in a dataset
 - D. To eliminate all biological variation
2. What does centering a variable accomplish?
 - A. Makes its variance equal to 1
 - B. Makes its mean equal to 0
 - C. Makes all observations positive
 - D. Makes its maximum equal to 1
3. Compared with Pareto scaling, autoscaling generally gives:
 - A. More influence to high-variance variables
 - B. Equal weighting to variables regardless of their original variance
 - C. No influence to low-variance variables
 - D. Identical results in all analyses
4. Why can autoscaling sometimes amplify measurement noise?
 - A. Variables with very small variance are divided by a small standard deviation.
 - B. Variables with large means are removed.

- C. Autoscaling changes categorical variables into continuous variables.
 - D. Autoscaling removes observations with extreme values.
5. Which statement best distinguishes normalization from scaling in the context of this course?
- A. They are always exactly the same operation.
 - B. Normalization often addresses systematic differences between samples, whereas scaling changes the relative weighting of variables.
 - C. Scaling operates only on samples, while normalization operates only on variables.
 - D. Normalization is required only for categorical variables.
6. What is a primary purpose of a log transformation for highly right-skewed biological measurements?
- A. Make every variable have variance 1
 - B. Reduce skewness and compress large values
 - C. Remove all outliers
 - D. Make all samples have identical distributions
7. In the simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

what does β_1 represent?

- A. The predicted value of Y when $X = 0$
 - B. The expected change in Y associated with a one-unit increase in X
 - C. The variance of Y
 - D. The residual for each observation
8. Ordinary least squares chooses regression coefficients to minimize:
- A. The sum of residuals
 - B. The sum of absolute predictor values
 - C. The sum of squared residuals
 - D. The variance of the predictor
9. In multiple linear regression, the coefficient β_j for X_j represents:
- A. The change in Y associated with a one-unit increase in X_j , holding the other predictors constant
 - B. The correlation between X_j and every other predictor
 - C. The causal effect of X_j on Y
 - D. The predicted value of Y when $X_j = 0$

10. A researcher finds a strong positive regression coefficient between a biomarker and disease severity. Which conclusion is most appropriate?
- A. The biomarker definitely causes disease severity.
 - B. The coefficient establishes causality because regression controls for error.
 - C. The biomarker is associated with disease severity, but the relationship is not necessarily causal.
 - D. The coefficient must be caused by measurement error.
11. Which situation is most characteristic of **overfitting**?
- A. High training error and high test error
 - B. Low training error and high test error
 - C. High training error and low test error
 - D. Equal training and test error
12. In logistic regression,
- $$P(Y = 1 \mid X)$$
- is:
- A. Restricted to either 0 or 1
 - B. A probability between 0 and 1
 - C. The log-odds
 - D. The odds ratio
13. If a logistic-regression coefficient $\beta_1 > 0$, increasing X_1 , while holding other predictors fixed, generally:
- A. Decreases the log-odds of $Y = 1$
 - B. Increases the log-odds of $Y = 1$
 - C. Has no effect on the odds
 - D. Makes $P(Y = 1)$ increase by exactly β_1
14. If $e^{\beta_1} = 2$, then a one-unit increase in X_1 , holding other predictors fixed:
- A. Doubles the probability of $Y = 1$
 - B. Adds 2 to the probability of $Y = 1$
 - C. Doubles the odds of $Y = 1$
 - D. Doubles the log-odds of $Y = 1$
15. Why can accuracy be misleading for a highly imbalanced classification problem?
- A. Accuracy cannot be calculated for binary classifiers.
 - B. A classifier can obtain high accuracy simply by predicting the majority class.

- C. Accuracy depends on the scale of the predictors.
 - D. Accuracy always decreases as class imbalance increases.
16. Which metric answers the question:
 “Of all truly positive samples, what fraction did the classifier correctly identify?”
- A. Precision
 - B. Specificity
 - C. Recall
 - D. Accuracy
17. An ROC curve plots:
- A. Precision versus recall
 - B. TPR versus FPR
 - C. Accuracy versus threshold
 - D. Specificity versus precision
18. When the positive class is rare and identifying positive cases is the main concern, which curve is often particularly informative?
- A. Residual plot
 - B. Precision-recall curve
 - C. Cost-function curve
 - D. Regression line
19. Compared with L2 regularization, L1 regularization has the distinctive ability to:
- A. Guarantee perfect test accuracy
 - B. Set some coefficients exactly to zero
 - C. Increase all coefficient magnitudes
 - D. Eliminate the need for validation
20. In scikit-learn logistic regression, decreasing C generally means:
- A. Weaker regularization
 - B. Stronger regularization
 - C. More training observations
 - D. A higher classification threshold

Part II. Conceptual Understanding

60 points

Answer each question clearly and concisely.

21. Preprocessing an Omics Dataset

10 points

A metabolomics dataset contains hundreds of metabolites. Some metabolites have concentrations thousands of times larger than others, and some samples have systematically higher overall signal intensities because different amounts of material were injected.

- (a) Explain why these two problems are conceptually different. (4 points)

- (b) Which type of preprocessing is primarily intended to address the difference in magnitude among variables: **scaling or normalization**? (2 points)

- (c) Which is primarily intended to address the systematic overall intensity differences among samples: **scaling or normalization**? (2 points)

- (d) Give one reason why autoscaling might be undesirable when some low-variance metabolites mostly contain measurement noise. (2 points)

22. Linear Regression Interpretation

10 points

Suppose a model predicting a biological response is

$$\hat{Y} = 10 + 3X_1 - 2X_2,$$

where X_1 is exposure level and X_2 is age.

- (a) Interpret the coefficient 3 for X_1 . (3 points)

- (b) Interpret the coefficient -2 for X_2 . (3 points)

- (c) Does the coefficient 3 prove that increasing exposure *causes* the biological response to increase? Explain. (2 points)

- (d) What is the interpretation of the intercept 10? Why might that interpretation sometimes have little practical meaning? **(2 points)**

23. Confounding

8 points

A researcher performs a simple linear regression and finds that people who exercise more have lower blood pressure.

After age is added to a multiple linear regression model, the coefficient for exercise becomes much smaller.

- (a) Give a plausible explanation for why the exercise coefficient changed after age was added. (4 points)

- (b) Does this necessarily mean that the first regression was mathematically incorrect? Explain. (2 points)

- (c) What general lesson does this illustrate about interpreting regression coefficients? (2 points)

24. Logistic Regression

8 points

Consider

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X.$$

Suppose $e^{\beta_1} = 3$.

- (a) What does the left side of the equation represent? (2 points)

- (b) Interpret $e^{\beta_1} = 3$. (3 points)

- (c) A student says:
“Every one-unit increase in X triples the probability of $Y = 1$.”
Is this correct? Explain. (3 points)

25. Classification Threshold

8 points

A disease-screening classifier produces $P(Y = 1 | X)$. You lower the classification threshold from 0.5 to 0.3.

- (a) What generally happens to recall? Why? (2 points)
- (b) What generally happens to the number of false positives? Why? (2 points)
- (c) What generally happens to precision? (2 points)
- (d) If missing a true disease case is extremely costly, explain why lowering the threshold may be reasonable even if precision decreases. (2 points)

26. Overfitting and Regularization

8 points

Two logistic-regression models are evaluated:

Model	Training Accuracy	Validation Accuracy
A	99%	72%
B	88%	85%

- (a) Which model shows stronger evidence of overfitting? Explain. (3 points)
- (b) Which model would you currently expect to generalize better? Explain. (2 points)
- (c) Explain how regularization could help Model A. (3 points)

27. Model Evaluation and Data Leakage

8 points

A student performs the following analysis:

1. Splits the complete dataset into development and test datasets.
2. Tries $C = 0.01, 0.1, 1, 10$, and 100.
3. Trains each model using the development data.
4. Evaluates all five models on the **test data**.
5. Selects the C with the highest test accuracy.
6. Reports that test accuracy as the model's final performance.

(a) What is wrong with this procedure? (3 points)

(b) What role should the test dataset have instead? (2 points)

(c) Describe how cross-validation could be used correctly to select C . (3 points)

End of Examination

Please make sure you have answered all questions.